

- Column 1: Row 2,3  $\rightarrow$  Dominoe 1    Row 4,5  $\rightarrow$  Dominoe 2  
 Row 6,7  $\rightarrow$  Dominoe 3  
 Column 2: Row 1,2  $\rightarrow$  Dominoe 4    Row 3,4  $\rightarrow$  Dominoe 5  
 Row 5,6  $\rightarrow$  Dominoe 6    Row 7,8  $\rightarrow$  Dominoe 7  
 Column 3: Dominoe 8 to 11 (Similar to Column 2)  
 Column 4: Dominoe 12 to 15 (Similar to Column 2)  
 Column 5: Dominoe 16 to 19 (Similar to Column 2)  
 Column 6: Dominoe 20 to 23 (Similar to Column 2)  
 Column 7: Dominoe 24 to 27 (Similar to Column 2)  
 Column 8: Dominoe 28 to 30 (Similar to Column 1)  $\square$

45. Prove that you can use dominoes to tile a rectangular checkerboard with an even number of squares.

**Solution:** Let, a rectangular checkerboard have  $m$  rows and  $n$  columns, where  $m \cdot n$  is even.

**Claim:** If  $m \cdot n$  is even, then at least one of  $m$  or  $n$  is even.

**Proof:** Let  $p: m \cdot n$  is even,  $q: m$  is even  $r: n$  is even

We have to prove that  $p \rightarrow (q \vee r)$  is True, by contraposition.

$\therefore$  We need to prove  $\neg(q \vee r) \rightarrow \neg p$  is True.

Let  $\neg(q \vee r)$  be True. By De Morgan's Law,  $(\neg q \wedge \neg r)$  is True. By defn. of AND,  $\neg q$  is True and  $\neg r$  is True.

$\neg p: m \cdot n$  is not even, i.e.  $m \cdot n$  is odd (Defn. 1)

$\neg q: m$  is not even, i.e.  $m$  is odd (Defn. 1)